

MEMORANDUM

Levelized cost of electricity: California utility solar versus combined-cycle gas

Subject: Whether any achievable set of conditions makes a California gas plant cheaper than utility-scale solar

Data: EIA-923 and EIA-860 plant-level operating data, NREL Annual Technology Baseline 2024, calendar year 2025

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The short version

Utility-scale solar is the lower-cost energy source in California. That much is already well established. The question worth asking is what would have to change for combined-cycle gas to win, and the answer is that nothing reachable does.

Solving for the value of each driver that would make the two plants tie produces a gas price half what plants actually pay, a cost of capital roughly double what developers face, a utilization rate above anything in the California fleet, and a **negative** carbon price. The result does not turn on any assumption I chose.

Plant	Capacity factor	LCOE 2025	LCOE 2030
	(2025)	(\$/MWh)	(\$/MWh)
Desert Quartzite (solar)	0.321	\$50.20	\$40.59
Gateway (combined cycle)	0.536	\$62.94	\$61.34
Solar advantage	—	\$12.75	\$20.76

Full-year 2025 operating data. LCOE in real 2022 dollars, pre-subsidy, 7% real discount rate over a 30-year life.

1. Two real plants, not two technologies

Rather than compare generic technology averages, this memo compares two operating plants in CAISO. Gateway Generating Station is a 619.7 MW combined cycle in Contra Costa County. Desert Quartzite is a 300 MW single-axis tracking solar farm in Riverside County.

Each plant's capacity factor and heat rate are derived from its actual 2025 output in EIA-923 rather than modeled. Gateway ran at a 0.536 capacity factor and burned 7.23 MMBtu per MWh. Desert Quartzite ran at 0.321, about 9% better than the national assumption NREL uses, which is what Riverside County sunlight buys. Capital and O&M costs come from the ATB, because plant-level construction cost is not publicly reported.

Gateway sits mid-fleet among California combined cycles, so the comparison is not built on an unusually good or bad performer.

2. Where gas loses is utilization, not capital cost

Gas and solar capital costs are nearly identical per kilowatt, at roughly \$1,495 and \$1,492. What separates them is how the fixed cost is spread. A plant that runs more hours divides its capital recovery across more megawatt-hours, so gas LCOE falls steeply with utilization and solar's does not move at all.

There is a floor underneath gas that solar does not have. Gateway's fuel and variable O&M come to **\$30.23 per megawatt-hour** before a dollar of construction cost is recovered, which is 60% of Desert Quartzite's entire levelized cost. That leaves roughly \$20 per megawatt-hour of headroom to recover a \$1,495 per kilowatt plant, and the only way to fit inside it is to run almost constantly.

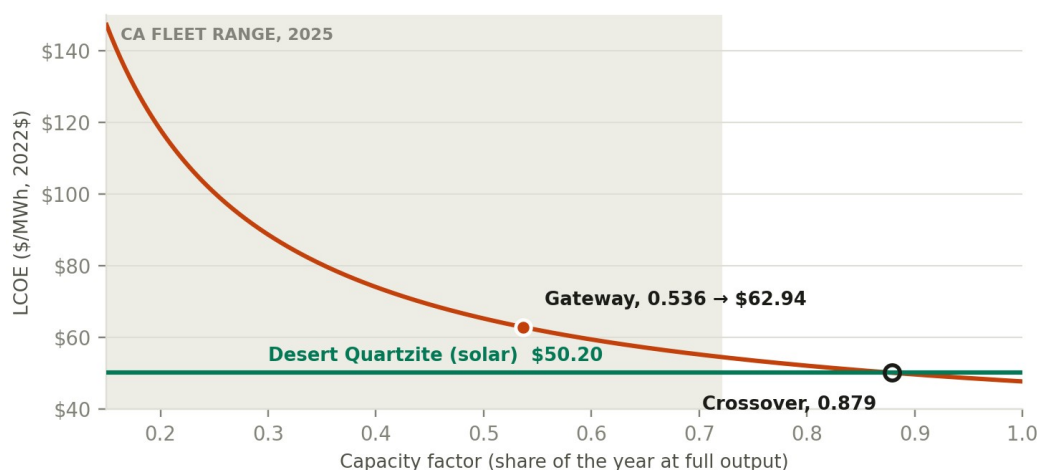


Figure 1. Gas LCOE against plant utilization, all other inputs at base case.

Gateway would need a capacity factor of **0.879** to match Desert Quartzite. Every other input can stay exactly where it is.

3. No California gas plant runs that hard

Across California's combined-cycle fleet, 2025 capacity factors ranged from 0.15 to 0.72. That spread alone swings gas LCOE by \$92.61 per megawatt-hour, nearly twice solar's entire levelized cost, which is why utilization rather than capital cost is the variable that decides gas economics.

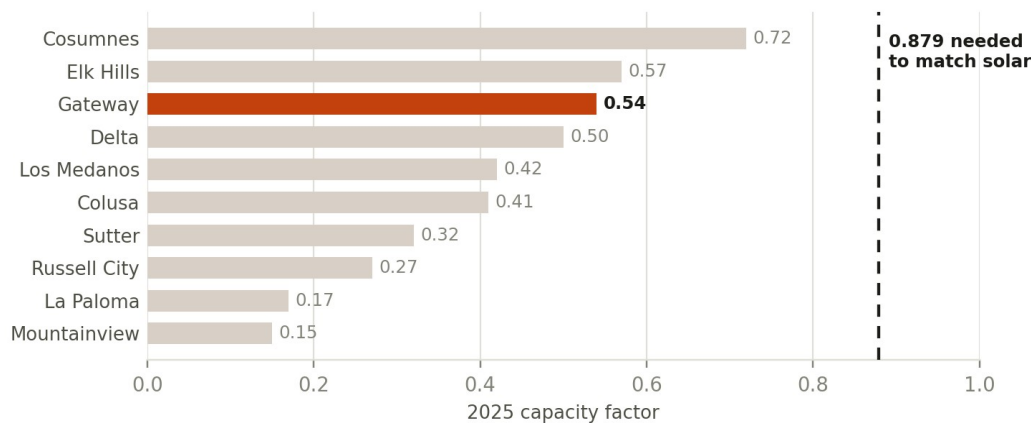


Figure 2. California combined-cycle capacity factors, 2025, against the utilization gas would need to match solar.

The hardest-working plant in the state reached 0.72, still well short of the 0.879 threshold. **The trend also runs the wrong way for gas:** as solar penetration rises, combined cycles are pushed further toward load-following, which lowers fleet utilization rather than raising it.

4. Nothing reachable reverses the result

Solving for each driver independently, holding the others at base case:

Driver	Value needed to tie	Actual, 2025
Delivered gas price	\$2.13/MMBtu	\$3.89/MMBtu
Real cost of capital	13.8%	7% assumed
Gas capacity factor	0.879	0.536 (fleet max 0.72)
CO ₂ allowance price	-\$33.24/tCO ₂	+\$32.48/tCO ₂

Goal Seek solutions setting the LCOE gap to zero. Gas prices in real 2022 dollars.

The carbon result is the informative one. Going in, the expectation was that California's cap-and-invest price might be what tips the balance. It is not. The balance was already tipped, and the \$32.48 August 2026 allowance price simply widens the gap from \$12.75 to **\$25.20** per megawatt-hour.

What this implies

1. **Gas is not competing on energy cost, and no policy lever under discussion changes that. Any case for new gas has to be made on dispatchability, not on price per megawatt-hour.**
2. **The binding constraints on California's transition are physical, not economic: firming, interconnection queues, and storage deployment. Those are where analysis and capital should go.**
3. **Falling fleet utilization is a compounding problem for existing gas. Each year solar displaces more midday output, gas capacity factors fall, and the per-megawatt-hour cost of the remaining fleet rises.**

Method and limitations

LCOE is computed as $(\text{CapEx} \times \text{FCR} + \text{fixed O\&M}) \div (\text{capacity factor} \times 8,760) + \text{variable O\&M} + \text{heat rate} \times \text{fuel price}$. The fixed charge rate is derived live from the discount rate as $r \div (1 - (1+r)^{-n})$ rather than assumed, which matters because the discount rate enters the model only through that term. Results are pre-subsidy and stated in real 2022 dollars; the EIA gas price is deflated from nominal 2025 using BLS CPI-U annual averages.

One methodological note is worth flagging because it moves a headline number. At a 7% real discount rate this model computes a fixed charge rate of 0.0806, against the 0.0566 implied by the ATB's own R&D Only financial case. Results therefore sit above NREL's published LCOE by design rather than by error, reflecting a private developer's cost of capital instead of a public-financing assumption.

Three limits belong on any read of these numbers. LCOE prices energy and nothing else, so it does not capture capacity value, curtailment, or the ability to ramp when the sun sets, which are the products gas actually sells. ATB gas capital costs predate the current turbine backlog, and contracts signed in 2025 and 2026 have come in materially higher, which biases this analysis toward gas looking cheap to build. And EIA 2025 figures are Early Release and not fully edited.

Sources: EIA Form 923 Schedules 2/3/4/5 and Form EIA-860, 2025 Early Release (30 June 2026); EIA series N3045CA3; NREL Annual Technology Baseline 2024 v4 workbook (revised 22 July 2026), Moderate scenario, R&D Only, 30-year capital recovery period; CARB cap-and-invest joint auction, August 2026; BLS CPI-U annual averages. Full Excel model available on request.